

u 2m



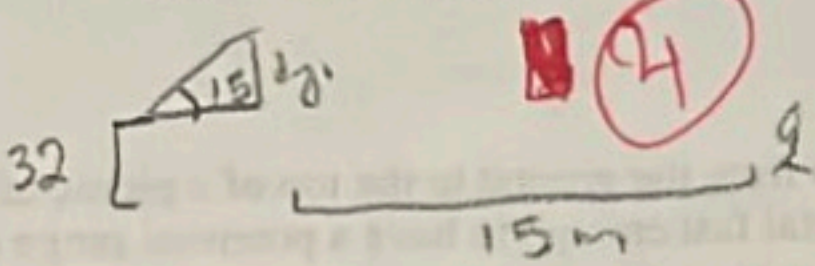
Chapter 1 Test- Kinematics

Rubric	Level 4+ 10 marks	Level 4 8-9 marks	Level 3 7 marks	Level 2 6 marks	Level 1 5 marks	Questions not answered or show little understanding of concepts.
Thinking/ Inquiry Questions answered using the formulas and units required to demonstrate understanding of the concepts.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of All concepts/ questions demonstrated.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of Most concepts/ questions demonstrated.	All questions answered using most of the correct formulas. A good understanding of most concepts/ questions demonstrated.	Most questions answered or attempted. Questions answered using some of the correct formulas. Some understanding of concepts/ questions demonstrated.	Some questions answered or attempted. Questions answered using limited formulas. Limited understanding of concepts/ questions demonstrated.	Questions not answered or show little understanding of concepts.

Clearly answer the following questions showing all your work. Round your final answer to 3 sd.

SUH
CAH
TOA

1. A quarterback throws a football at 15° above the horizontal from the edge of a 32.0 m high cliff with a constant horizontal velocity of 22 m/s. As the ball is thrown, a player at the bottom of the cliff 15 m away from its base, accelerates at 3.1 m/s^2 to try and catch the ball. Show by calculation if it is possible for the player to catch the ball.



$v_x = 22 \text{ m/s}$
 $v_y: \tan 15^\circ = \frac{v_y}{22 \text{ m/s}}$
 $v_y = 5.89 \text{ m/s}$

$y = -9.8 \text{ m/s}^2$
 $\Delta y = -32 \text{ m}$
 person: $d = 15 \text{ m}$
 $a = 3.1 \text{ m/s}^2$
 $v_i = 0 \text{ m/s}$

1. Time of football

$\Delta y = v_{iy}t + \frac{1}{2}at^2$
 $-32 \text{ m} = (5.89 \text{ m/s})t + \frac{1}{2}(-9.8 \text{ m/s}^2)t^2$
 $-32 \text{ m} = 5.89 \text{ m/s}t - 4.9 \text{ m/s}^2t^2$
 $4.9t^2 - 5.89t - 32 = 0$

$t = \frac{-(-5.89) \pm \sqrt{(-5.89)^2 - 4(4.9)(-32)}}{2(4.9)}$

$t = 3.23 \text{ s} \text{ or } -2.02 \text{ s}$

2. How far it travels

$v = \frac{\Delta x}{t}$
 $\Delta x = v \cdot t$
 $\Delta x = (22 \text{ m/s})(3.23 \text{ s})$
 $\Delta x = 71.06 \text{ m}$

$\Delta x = v_{ix}t + \frac{1}{2}at^2$
 $= (22 \text{ m/s}) + \frac{1}{2}(3.1 \text{ m/s}^2)(3.23 \text{ s})^2$
 $\Delta x_p = 83.4 \text{ m}$

person: $83.4 \text{ m} + 15 \text{ m}$
 $\text{person} = 98.4 \text{ m}$

3. Person's time

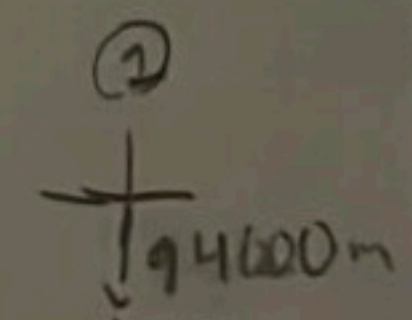
$\Delta x = v_{ix}t + \frac{1}{2}at^2$
 $15 \text{ m} = \frac{1}{2}(3.1 \text{ m/s}^2)t^2$
 $15 \text{ m} = (1.55 \text{ m/s}^2)t^2$
 $t = 3.11 \text{ s}$

person misses
 the ball by run
 too far 98.4 m

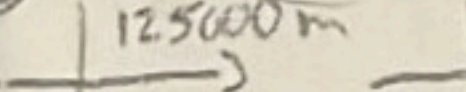
2. A cruise ship sails 94.0 km [S], then 125 km [E] and finally 85.0 km [E 45 N]. The trip takes 4.25 hours to complete.

- a. Calculate the average velocity of the sailboat.

- b. What is the acceleration from the first leg of the trip to the second leg of the trip if the transition took 180 s?



2. took 180 s?



$\Delta R^2 = \Delta x^2 + \Delta y^2$

$\Delta R = \sqrt{(185104 \text{ m})^2 + (33896 \text{ m})^2}$

$\Delta R = 185135 \text{ m} \text{ [E } 10^\circ \text{ S]}$

$\Delta x = 185104 \text{ m} \text{ [E]}$

$b. a = \frac{\Delta v}{t}$

$= \frac{12.1 \text{ m/s} \text{ [E } 10^\circ \text{ S]}}{140 \text{ s}}$

$a = 0.07 \text{ m/s}^2 \text{ [E } 10^\circ \text{ S]}$

$V = \frac{\Delta}{t}$

$V = \frac{185135 \text{ m} \text{ [E } 10^\circ \text{ S]}}{15300 \text{ s}}$

$V = 12.1 \text{ m/s} \text{ [E } 10^\circ \text{ S]}$

$a. V = 12.1 \text{ m/s} \text{ [E } 10^\circ \text{ S]}$

Jump:

$\Delta y_1 = 94000 \text{ m} \text{ [S]}$

$\sin 45 = \frac{\Delta y_2}{85000 \text{ m}}$

$\Delta y_2 = 60104 \text{ m} \text{ [N]}$

$\Delta y_3 = 33896 \text{ m} \text{ [S]}$

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ou 2m

$$M: v_i = 1.2 \text{ m/s}$$

$$a = ?$$

$$v_f = ?$$

$$d = 0.38 \text{ km}$$

$$C: v = 12 \text{ m/s}$$

3. Christoffer is biking at a constant velocity of 12 m/s on Shaftsbury. Matthew is walking home in the same direction at 1.2 m/s. When Matthew is passed by Christoffer he begins to accelerate (constant acceleration) to a run to try and catch up. What acceleration must Matthew maintain and what will his final velocity be when he finally catches Christoffer at Bathurst 0.38 km later? (assume Christoffer and Matthew are agile enough to elude contact with other students/cars without slowing down).

4. Patrick continues his crazy rollerblading ways and tries to jump from the ground to the top of a picnic table that is 1.2 m tall. He jumps at an angle of 70° above the horizontal fast enough to have a potential range of 5m in 1.7s (as long as he doesn't hit a table...). The table is 2.5 m away...

- a. Does he land on top of the table?

$$v_i = \frac{5\text{m}}{1.7\text{s}}$$

$$v_{iy} = 2.76 \text{ m/s}$$

$$v_{ix} = 1.01 \text{ m/s}$$

$$v_i = 2.94 \text{ m/s}$$

$$v_{iy} = 2.76 \text{ m/s}$$

$$\Delta d = v_{ix} t + \frac{1}{2} a t^2$$

$$d_p = (1.01 \text{ m/s})(0.855 \text{ s})$$

$$d_p = 0.85 \text{ m}$$

yes, he lands on top of the table $d_p = 4.35 \text{ m}$

$$d = 0.855 \text{ m} = 0.29 \times$$

on sep

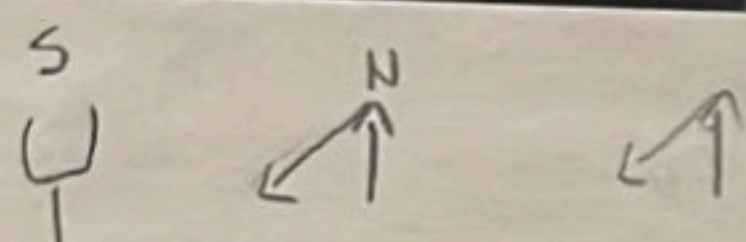
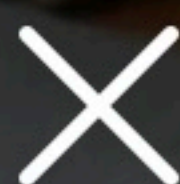
slat of apr

- b. Lauren is 1.90m tall with her Moonshoes on, and is standing 85 cm away from Patrick... does he clear her head?

No, he'll hit the ground

work done on sep slat

u 48s



$$d=0$$

$$v = \frac{d}{t} \rightarrow$$

$$h \quad v_0 \quad x = \text{range} \quad y =$$

$$V = \frac{d}{t}$$

$$v^2 = v_0^2 + 2ad$$

$$\frac{d}{t} = v_0 + 2ax$$

$$4. \quad v_{ix} = 2.94 \text{ m/s}$$

$$v_{iy} = 8.08 \text{ m/s}$$

$$\Delta dy = v_{iy} t + \frac{1}{2} a t^2$$

$$1.2 = 8.08 t - 4.9 t^2$$

$$4.9 t^2 - 8.08 t + 1.2 = 0$$

$$t = 1.485 / 0.16 \times \text{Quadratic formula}$$

$$b. \quad \begin{matrix} p \\ q \end{matrix}$$

$$\begin{matrix} l \\ r \end{matrix} \quad 1.90 \text{ m}$$

$$0.85 \text{ m}$$

$$dy = v_{iy} t + \frac{1}{2} a t^2$$

$$dy = (8.08 \text{ m/s}) t + \frac{1}{2} (-9.8 \text{ m/s}^2) (1.36 \text{ s})^2$$

$$dy = -0.98 \text{ m}$$

$$\Delta d_p = v_{ix} t + \frac{1}{2} a t^2$$

$$d_p = (2.94 \text{ m/s})(1.485 \text{ s})$$

$$d_p = 4.35 \text{ m}$$

X

$$dy = v_{iy} t + \frac{1}{2} a t^2$$

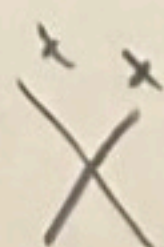
$$1.90 = 8.08 t - 4.9 t^2$$

$$t = 0.245 / 1.36 \text{ s}$$

X

in the same
at acceleration)
nal velocity be
Matthew are agile

the top of a picnic table
a potential range of



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u 28s



APPLICATION – 10 marks

Name

Answer the following question clearly showing your work and submit it to the appropriate Assignment Drop Box. Round your final answer to 3 sd.

The following rubric will be used to assess short answer questions.

Rubric	Level 4+ 10 marks	Level 4 8-9 marks	Level 3 7 marks	Level 2 6 marks	Level 1 5 mark	NOT YET
Application Questions answered using the formulas and units required to demonstrate understanding of the concepts.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of All concepts/questions demonstrated.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of Most concepts/questions demonstrated.	All questions answered using most of the correct formulas. A good understanding of most concepts/questions demonstrated.	Most questions answered or attempted. Questions answered using some of the correct formulas. Some understanding of concepts/questions demonstrated.	Some questions answered or attempted. Questions answered using limited formulas. Limited understanding of concepts/questions demonstrated.	Questions not answered or show little understanding of concepts.

1. Assuming the initial velocity of an arrow released during our projectile lab was 27.5 m/s at an angle of 25 degrees above the horizon,

- a. How high would the arrow have to be launched from the ground to land 80m away from the launch point?

$$v_i = 27.5 \text{ m/s}$$

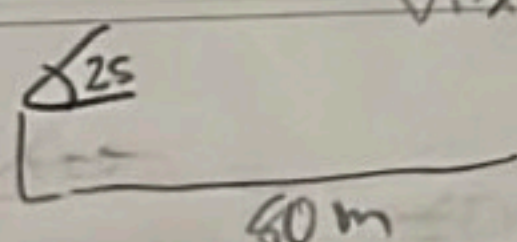
$$v_{iy} = 11.6 \text{ m/s}$$

$$v_{ix} = 24.9 \text{ m/s}$$

$$dy = v_{iy}t + \frac{1}{2}at^2$$

$$dy = 11.6 \text{ m/s} (3.21 \text{ s}) + \frac{1}{2}(-9.8 \text{ m/s}^2)(3.21 \text{ s})^2$$

$$dy = -13.3 \text{ m}$$



$$v = \frac{d}{t}$$

$$t = \frac{80 \text{ m}}{24.9 \text{ m/s}}$$

$$t = 3.21 \text{ s}$$

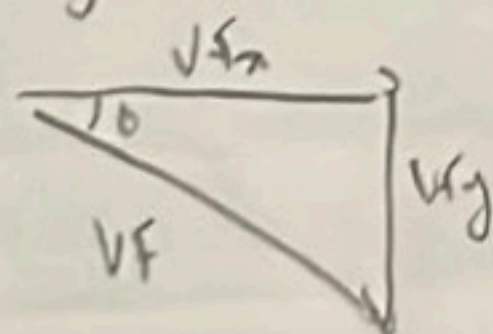
- b. Calculate the final velocity of the arrow just before it hits the ground and explain the difference compared to the initial velocity.

$$v_{fx} = 24.9 \text{ m/s [F]}$$

$$v_{fy}^2 = v_{iy}^2 + 2a\Delta y$$

$$v_{fy} = \sqrt{(11.6 \text{ m/s})^2 + 2(-9.8 \text{ m/s}^2)(-13.3 \text{ m})}$$

$$v_{fy} = 19.9 \text{ m/s [D]}$$



$$v_f^2 = v_{fx}^2 + v_{fy}^2$$

$$v_f = \sqrt{(24.9 \text{ m/s [F]})^2 + (19.9 \text{ m/s [D]})^2}$$

$$v_f = 31.9 \text{ m/s}$$

$$\tan \theta = \frac{v_{fy}}{v_{fx}}$$

$$\tan \theta = \frac{19.9 \text{ m/s}}{24.9 \text{ m/s}}$$

$$\theta = 39^\circ$$

$\therefore v_f = 31.9 \text{ m/s [F } 39^\circ \text{ D]}$ is greater than v_i bc of acceleration due to gravity



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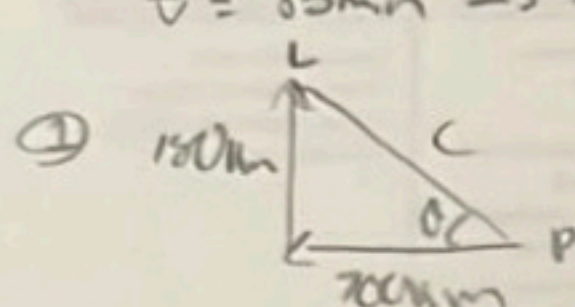
u 15s



2. An airplane takes off from London, England at 11:05 am and is scheduled to arrive at Prague, Czech Republic at 12:30 pm. The pilot is alerted to a strong wind of 78 km/h [N 35°W].

London is 180 km North and 700 km west of Prague. If the wind velocity does not change the entire trip, determine the airspeed and heading that the pilot must maintain in order to reach the destination on time.

$$t = 85 \text{ min} \rightarrow 5100 \text{ s}$$



$$C^2 = a^2 + b^2$$

$$C = \sqrt{(180 \text{ km})^2 + (700 \text{ km})^2}$$

$$C = 722.77 \text{ km}$$

$$\rightarrow 722770 \text{ m}$$

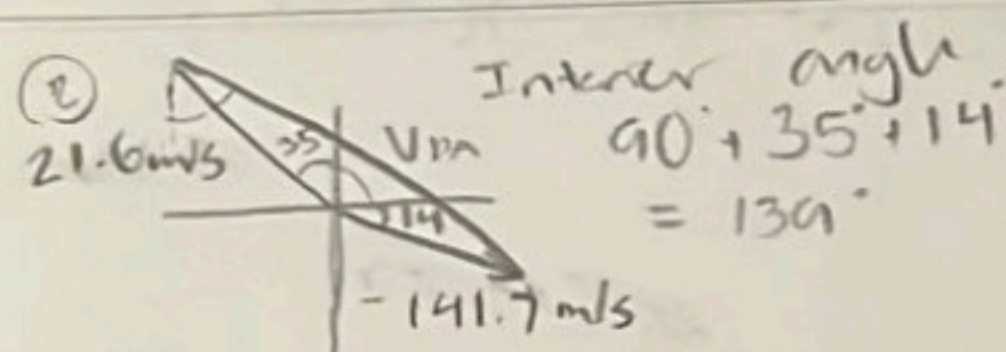
$$\tan \theta = \frac{180 \text{ km}}{700 \text{ km}}$$

$$\theta = 14^\circ$$

velocity it needs to travel at:

$$d = \frac{v}{t}$$

$$v_{\text{req}} = \frac{722770 \text{ m}}{5100 \text{ s}} = 141.7 \text{ m/s [E 14° S]}$$



Find velocity of v_{pa} (Airspeed)

$$v_{\text{pa}}^2 = a^2 + b^2 - 2ab \cos \theta$$

$$v_{\text{pa}} = \sqrt{(21.6 \text{ m/s})^2 + (141.7 \text{ m/s})^2 - 2(21.6 \text{ m/s})(141.7 \text{ m/s}) \cos 139^\circ}$$

$$v_{\text{pa}} = 158.6 \text{ m/s}$$

③ Find angle.

$$\frac{\sin \theta}{141.7 \text{ m/s}} = \frac{\sin 139^\circ}{158.6 \text{ m/s}}$$

$$\sin \theta = 0.586$$

$$\theta = \sin^{-1}(0.586)$$

$$\theta = 36^\circ$$

$$\theta = 35^\circ + 36^\circ$$

$$\theta = 71^\circ$$

$\therefore$ airspeed and heading that needs to be maintained is 159 m/s [S 71° E]



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